Distributions and Maximum Likelihood

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$? Show your complete algebraic work.

Final Answer: $\frac{3}{10}$. The probability mass function is given by $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. We need to find $P(X \leq 2)$. This is the sum of probabilities for $X = 1$ and $X = 2$. $P(X \leq 2) = P(X = 1) + P(X = 2)$. $P(X = 1) = \frac{1}{10}$ and $P(X = 2) = \frac{2}{10}$. Therefore, $P(X \leq 2) = \frac{1}{10} + \frac{2}{10} = \frac{3}{10}$.

2 **[Foundational]** • **Target: Discrete Random Variables and Bernoulli Likelihood**

A dataset consists of observations $x_1, x_2, \ldots, x_n$ that are assumed to be drawn independently from a Bernoulli distribution with parameter p. The probability mass function is given by $P(X = k; p) = p^k(1 - p)^{1-k}$ for $k \in \{0, 1\}$. The following bar chart shows the frequency distribution of these observations, where $n = 50$. { "type": "bar", "labels": ["0", "1"], "data": [20, 30], "title": "Frequency Distribution of Bernoulli Observations" } Using the method of Maximum Likelihood Estimation (MLE), determine the estimated value of p for this distribution.

Final Answer: $\frac{3}{5}$. The Bernoulli distribution has the probability mass function $P(X = k; p) = p^k(1 - p)^{1-k}$. The likelihood function for n independent observations is $L(p; x_1, \ldots, x_n) = \prod_{i=1}^{n} p^{x_i}(1 - p)^{1-x_i}$. This can be rewritten as $L(p) = p^{\sum x_i}(1 - p)^{n - \sum x_i}$. The log-likelihood function is $\ln L(p) = (\sum x_i)\ln p + (n - \sum x_i)\ln(1 - p)$. To find the maximum likelihood estimate (MLE) of p, we take the derivative of the log-likelihood with respect to p.

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3 [Foundational] • Target: Continuous Gaussian Normal Distributions

Consider a sequence of n independent Bernoulli trials, where the probability of success in each trial is p. Let X be the random variable representing the number of successes in these n trials. Thus, $X \sim \text{Binomial}(n, p)$. We are given a dataset of N observed values $\{x_1, x_2, \ldots, x_N\}$ for X, where each x_i is an observed count of successes from n trials. The observed data are: [3, 4, 2, 5, 4, 3, 2, 4, 5, 3]. If we assume that $n = 5$ (the number of trials for each observation is fixed at 5), find the value of p that maximizes the likelihood of observing this dataset. Then, compute the maximum likelihood estimate (MLE) of the probability of success, p, and round your answer to two decimal places.

Final Answer: 0.70. The random variable X follows a Binomial distribution: $X \sim \text{Binomial}(n, p)$. The probability mass function (PMF) of the Binomial distribution is given by $P(X = k|n, p) = \binom{n}{k} p^k (1-p)^{n-k}$, where k is the number of successes. We are given $n = 5$ and a dataset of $N = 10$ observations: $\{x_1, x_2, \ldots, x_{10}\} = \{3, 4, 2, 5, 4, 3, 2, 4, 5, 3\}$. The likelihood function for this dataset, given p, is the product of the PMFs for each observation.

4 [Foundational] • Target: The Likelihood Function and Log-Likelihood Formulation

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$. Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ. Part 2: Set the derivative to zero to find the MLE of λ.

Final Answer: $\hat{\lambda} = \frac{3+5+4}{3} = 4$. The log-likelihood function for $n = 3$ observations is $\ell(\lambda) = (x_1 \log(\lambda) - \log(x_1!) - \lambda) + (x_2 \log(\lambda) - \log(x_2!) - \lambda) + (x_3 \log(\lambda) - \log(x_3!) - \lambda)$. Part 1: The derivative of the log-likelihood function with respect to λ is.

5 [Foundational] • Target: Properties of Estimators: Bias and Consistency

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2? Show your complete algebraic work.

Final Answer: 0.36. The random variable X follows a binomial distribution, as it represents the number of successes in a fixed number of independent Bernoulli trials. The expected value of a binomial random variable is given by $E[X] = np$, where n is the number of trials and p is the probability of success. We are given $E[X] = 18$ and $p = 0.6$. Therefore, $np = 18$ implies $n(0.6) = 18$. Solving for n, we get $n = \frac{18}{0.6} = 30$. So, there are 30 trials. The variance of a binomial random variable i.

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6 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Consider a dataset of observations $x_1, x_2, \ldots, x_n$ assumed to be drawn independently from an exponential distribution with probability density function (PDF) $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$ and $\lambda > 0$. The likelihood function for this dataset is given by $L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$. To find the Maximum Likelihood Estimator (MLE) for λ, we first consider the log-likelihood function $\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$. If the observed data points are $\{1.2, 2.5, 0.8, 3.1, 1.5 \backslash)$, and $n = 5$, determine the MLE of λ.

Final Answer: $\frac{5}{9.1}$. The log-likelihood function is $\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$. To find the MLE, we differentiate $\ln L(\lambda)$ with respect to λ and set the derivative to zero: $\frac{d}{d\lambda} \ln L(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i$ Setting this to zero: $\frac{n}{\lambda_{MLE}} - \sum_{i=1}^{n} x_i = 0$ $\frac{n}{\lambda_{MLE}} = \sum_{i=1}^{n} x_i$.

7 [Foundational] • Target: Discrete Random Variables and Bernoulli Likelihood

A dataset consists of $n = 100$ independent and identically distributed (i.i.d.) observations $x_1, x_2, \ldots, x_{100}$. These observations are drawn from a Bernoulli distribution with parameter p. The probability mass function (PMF) of a Bernoulli distribution is given by $P(X = x; p) = p^x (1-p)^{1-x}$ for $x \in \{0, 1\}$. The likelihood function $L(p)$ for this dataset is the product of the PMFs of the individual observations. If we observe $k = 60$ successes (i.e., $x_i = 1$) and $n - k = 40$ failures (i.e., $x_i = 0$), what is the likelihood function $L(p)$ for p? Show your complete algebraic work.

Final Answer: $p^{60}(1-p)^{40}$. The likelihood function $L(p)$ is the product of the probabilities of observing each data point given the parameter p. For i.i.d. Bernoulli trials, $L(p) = \prod_{i=1}^{n} P(X = x_i; p)$. Given $n = 100$ observations, with $k = 60$ successes (where $x_i = 1$) and $n - k = 40$ failures (where $x_i = 0$), the likelihood function can be written as: $L(p) = \left(\prod_{i:x_i=1} p^1 (1-p)^{1-1} \right) \times \left(\prod_{i:x_i=0} p^0 (1-p)^{1-0} \right)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Continuous Gaussian Normal Distributions

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k? Show your complete algebraic work.

Final Answer: 1/10. The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{4} P(X = x) = 1$. Substituting the given probability mass function, we have: $P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$ $k(1) + k(2) + k(3) + k(4) = 1$ Factoring out k: $k(1 + 2 + 3 + 4) = 1$ $k(10) = 1$ Solving for k: $k = \frac{1}{10}$. Common algebraic traps include incorrectly summing the values of x or forgetting to set the sum of probabilities to 1. For example, setting.

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9 [Intermediate] • Target: The Likelihood Function and Log-Likelihood Formulation

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$? Show your complete algebraic work.

Final Answer: 0.50. For a continuous uniform distribution over the interval $[a, b]$, the probability density function (PDF) is $f(x) = \frac{1}{b-a}$ for $a \le x \le b$, and 0 otherwise. In this case, $a = 2$ and $b = 8$, so the length of the interval is $b - a = 8 - 2 = 6$. The PDF is $f(x) = \frac{1}{6}$ for $2 \le x \le 8$. The probability $P(4 < X < 7)$ is the integral of the PDF from 4 to 7, or equivalently, the area under the PDF curve between 4 and 7. $P(4 < X < 7) = \int_4^7 \frac{1}{6} dx$ Since the PDF.

10 [Intermediate] • Target: Properties of Estimators: Bias and Consistency

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$? Show your complete algebraic work.

Final Answer: $\frac{5}{8}$. The problem asks for the probability that the random variable X (number of heads in 3 coin tosses) is less than 2. This means we need to find $P(X < 2)$. The possible values of X that are less than 2 are $X = 0$ and $X = 1$. Therefore, $P(X < 2) = P(X = 0) + P(X = 1)$. From the given probability distribution: $P(X = 0) = \frac{1}{8}$ $P(X = 1) = \frac{3}{8}$ So, $P(X < 2) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$. Simplifying the fraction, $P(X < 2) = \frac{1}{2}$. Looking at the choices provi.

11 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let X be the first integer generated and Y be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

Final Answer: 1/3. The problem states that the probability of generating an integer in a range $[a, b]$ is proportional to $b - a$ for $0 \le a \le b \le 1$. This describes a continuous uniform distribution on the interval $[0, 1]$. Therefore, the probability density function (PDF) for both X and Y is $f(x) = 1$ for $0 \le x \le 1$ and 0 otherwise. We are interested in the probability $P(X + Y < 1)$. Since X and Y are independent and both follow a $U(0, 1)$ distribution, their joint PDF is.

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12 **[Intermediate]** • **Target: Discrete Random Variables and Bernoulli Likelihood**

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)? Show your complete algebraic work.

Final Answer: $\frac{1}{3}$. The random variable X can take on the values 1, 2, and 3. The probability distribution is given by $P(X = x) = kx$, where k is a constant. The sum of the probabilities for all possible values of X must equal 1. $P(X = 1) + P(X = 2) + P(X = 3) = 1$ Substituting the given formula for the probabilities: $k(1) + k(2) + k(3) = 1$ $k + 2k + 3k = 1$ $6k = 1$ Solving for k: $k = \frac{1}{6}$ Now we can find the probability $P(X = 2)$: $P(X = 2) = k(2)$ Substitute the value of k we found.

13 **[Intermediate]** • **Target: Continuous Gaussian Normal Distributions**

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario? Show your complete algebraic work.

Final Answer: $\lambda = 3$. The problem statement explicitly defines that the number of defective items, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The rate parameter λ is a direct characteristic of the Poisson distribution that determines its mean and variance. Therefore, the value of λ is directly given as 3.

14 **[Intermediate]** • **Target: The Likelihood Function and Log-Likelihood Formulation**

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{36}{12} = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Properties of Estimators: Bias and Consistency** • **Cross-Lesson: 6.01 Linear Regression and OLS**

Given $n = 16$ independent observations from a normal distribution with sum $\sum_{i=1}^{16} x_i = 48$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{48}{16} = 3$.

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16 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Given $n = 20$ independent observations from a normal distribution with sum $\sum_{i=1}^{20} x_i = 60$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{60}{20} = 3$.

17 [Challenging] • Target: Discrete Random Variables and Bernoulli Likelihood • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Given $n = 24$ independent observations from a normal distribution with sum $\sum_{i=1}^{24} x_i = 72$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{72}{24} = 3$.

18 [Challenging] • Target: Continuous Gaussian Normal Distributions • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{12}{4} = 3$.

19 [Challenging] • Target: The Likelihood Function and Log-Likelihood Formulation • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Given $n = 8$ independent observations from a normal distribution with sum $\sum_{i=1}^{8} x_i = 24$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$? Show your complete algebraic work.

Final Answer: $\hat{\mu}_{\text{MLE}} = 3$. The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{24}{8} = 3$.

20 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 3.01 Gradient Descent and Optimization

Expected value E(X) for discrete X:.

Final Answer: $\sum x_i P(x_i)$. Applying Unit 3 · Probability, Random Variables & Distributions principles, Weighted average of outcomes. Therefore, the verified correct option is $\sum x_i P(x_i)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(5, 4)$. What is the mean μ and variance σ^2 of this feature?

a. **$\mu = 5, \; \sigma^2 = 4$** ✓

b. $\mu = 4, \; \sigma^2 = 5$

c. $\mu = 5, \; \sigma^2 = 2$

d. $\mu = 2, \; \sigma^2 = 4$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 5$ and variance $\sigma^2 = 4$ (with standard deviation $\sigma = 2$).

22 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(10, 9)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 10, \; \sigma^2 = 3$

b. $\mu = 9, \; \sigma^2 = 10$

c. **$\mu = 10, \; \sigma^2 = 9$** ✓

d. $\mu = 3, \; \sigma^2 = 9$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 10$ and variance $\sigma^2 = 9$ (with standard deviation $\sigma = 3$).

23 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(15, 16)$. What is the mean μ and variance σ^2 of this feature?

a. **$\mu = 15, \; \sigma^2 = 16$** ✓

b. $\mu = 16, \; \sigma^2 = 15$

c. $\mu = 15, \; \sigma^2 = 4$

d. $\mu = 4, \; \sigma^2 = 16$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 15$ and variance $\sigma^2 = 16$ (with standard deviation $\sigma = 4$).

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24 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(20, 25)$. What is the mean μ and variance σ^2 of this feature?

a. **$\mu = 20, \; \sigma^2 = 25$** ✓

b. $\mu = 25, \; \sigma^2 = 20$

c. $\mu = 20, \; \sigma^2 = 5$

d. $\mu = 5, \; \sigma^2 = 25$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 20$ and variance $\sigma^2 = 25$ (with standard deviation $\sigma = 5$).

25 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

a. **$\mu = 25, \; \sigma^2 = 36$** ✓

b. $\mu = 36, \; \sigma^2 = 25$

c. $\mu = 25, \; \sigma^2 = 6$

d. $\mu = 6, \; \sigma^2 = 36$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 25$ and variance $\sigma^2 = 36$ (with standard deviation $\sigma = 6$).

26 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(30, 49)$. What is the mean μ and variance σ^2 of this feature?

a. **$\mu = 30, \; \sigma^2 = 49$** ✓

b. $\mu = 49, \; \sigma^2 = 30$

c. $\mu = 30, \; \sigma^2 = 7$

d. $\mu = 7, \; \sigma^2 = 49$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 30$ and variance $\sigma^2 = 49$ (with standard deviation $\sigma = 7$).

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27 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Given \(n = 4 \) independent observations from a normal distribution with sum \(\sum_{i=1}^{4} x_i = 12 \), what is the Maximum Likelihood Estimate (MLE) \(\hat{\mu}_{\text{MLE}} \)?

a. \(\hat{\mu}_{\text{MLE}} = 3 \) ✓

b. \(\hat{\mu}_{\text{MLE}} = 6 \)

c. \(\hat{\mu}_{\text{MLE}} = 12 \)

d. \(\hat{\mu}_{\text{MLE}} = 1.5 \)

The log-likelihood \(\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum (x_i - \mu)^2 \). Setting \(\frac{\partial \ln L}{\partial \mu} = 0 \) yields the sample mean \(\hat{\mu} = \frac{1}{n}\sum x_i = \frac{12}{4} = 3 \).

28 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Given \(n = 8 \) independent observations from a normal distribution with sum \(\sum_{i=1}^{8} x_i = 24 \), what is the Maximum Likelihood Estimate (MLE) \(\hat{\mu}_{\text{MLE}} \)?

a. \(\hat{\mu}_{\text{MLE}} = 3 \) ✓

b. \(\hat{\mu}_{\text{MLE}} = 6 \)

c. \(\hat{\mu}_{\text{MLE}} = 24 \)

d. \(\hat{\mu}_{\text{MLE}} = 1.5 \)

The log-likelihood \(\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum (x_i - \mu)^2 \). Setting \(\frac{\partial \ln L}{\partial \mu} = 0 \) yields the sample mean \(\hat{\mu} = \frac{1}{n}\sum x_i = \frac{24}{8} = 3 \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

In a binary click-through model, an ad is clicked $k = 40$ times out of $n = 200$ impressions. What is the Maximum Likelihood Estimate $\hat{p}_{MLE}$ of the click probability?

a. 0.20 ✓

b. 0.25

c. 0.16

d. 0.40

For Bernoulli trials, $\hat{p}_{MLE} = k/n = 40/200 = 0.20$.

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30 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

What is the variance of a Bernoulli random variable with success probability $p = 0.3$?

a. 0.21 ✓

b. 0.30

c. 0.70

d. 0.49

$\text{Var}(X) = p(1-p) = 0.3(0.7) = 0.21$.

31 [Intermediate] • Target: Discrete Random Variables and Bernoulli Likelihood

Show how maximizing the Bernoulli log-likelihood relates to the binary cross-entropy loss function used in neural networks.

a. **Maximizing Bernoulli log-likelihood is mathematically identical to minimizing binary cross-entropy** ✓

b. Cross-entropy is the reciprocal of the log-likelihood

c. They only match when $p = 0.5$

d. Cross-entropy ignores negative classes

Log-likelihood is $\sum[y_i \ln p + (1-y_i)\ln(1-p)]$. Binary cross-entropy is the negative log-likelihood $-\text{LL}$.

32 [Intermediate] • Target: Continuous Gaussian Normal Distributions

A feature vector component follows $X \sim \mathcal{N}(50, 16)$. What are its mean μ and standard deviation σ?

a. $\mu = 50, \quad \sigma = 4$ ✓

b. $\mu = 50, \quad \sigma = 16$

c. $\mu = 16, \quad \sigma = 50$

d. $\mu = 25, \quad \sigma = 4$

Notation $\mathcal{N}(\mu, \sigma^2) \Rightarrow \mu = 50$, $\sigma^2 = 16 \Rightarrow \sigma = 4$.

33 [Intermediate] • Target: Continuous Gaussian Normal Distributions

By the empirical $(68-95-99.7)$ rule, approximately what percentage of samples from $\mathcal{N}(\mu, \sigma^2)$ fall within the interval $(\mu - 2\sigma, \mu + 2\sigma)$?

a. 95.4% ✓

b. 68.2%

c. 99.7%

d. 50.0%

Approximately 95.4% of Gaussian mass lies within ± 2 standard deviations.

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34 [Intermediate] • Target: Continuous Gaussian Normal Distributions

Standardize the value $x = 62$ from the distribution $\mathcal{N}(50, 16)$ into a standard normal z-score.

a. $z = 3.0$ ✓

b. $z = 0.75$

c. $z = 12$

d. $z = 1.5$

$z = \frac{x - \mu}{\sigma} = \frac{62 - 50}{4} = \frac{12}{4} = 3.0.$

35 [Intermediate] • Target: The Likelihood Function and Log-Likelihood Formulation

Why is the log-likelihood $\ln L(\theta) = \sum_{i=1}^{n} \ln p(x_i|\theta)$ preferred over raw likelihood in machine learning optimization?

a. **It transforms products of probabilities into sums, preventing numerical underflow and simplifying derivatives** ✓

b. It converts non-convex problems into concave problems unconditionally

c. It eliminates the need for normalization

d. It ensures the sample mean is zero

Multiplying many small probabilities causes float underflow; summing logs is numerically stable and derivatives split cleanly.

36 [Intermediate] • Target: The Likelihood Function and Log-Likelihood Formulation

For n independent samples from $\mathcal{N}(\mu, \sigma^2)$, what is the analytical expression for the log-likelihood?

a. $-\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$ ✓

b. $-\frac{n}{2}\ln(2\pi) - \sum_{i=1}^{n}(x_i - \mu)$

c. $\frac{n}{2\sigma^2}\sum(x_i - \mu)^2$

d. $-\frac{1}{\sigma^2}\sum(x_i - \mu)$

Taking natural log of $\prod \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$ gives this standard form.

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37 [Intermediate] • Target: The Likelihood Function and Log-Likelihood Formulation

Given 5 observations $\{2, 4, 4, 6, 9\}$, compute the sample mean $\bar{x}$, which is the Gaussian MLE $\hat{\mu}_{MLE}$.

a. 5.0 ✓

b. 4.0

c. 4.6

d. 5.5

$\bar{x} = \frac{2+4+4+6+9}{5} = \frac{25}{5} = 5.0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Why is the Maximum Likelihood Estimator for variance $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \bar{x})^2$ biased for finite samples?

a. **Its expected value is $\frac{n-1}{n}\sigma^2 < \sigma^2$, underestimating true variance due to using sample mean $\bar{x}$** ✓

b. It only works for standard normal data

c. Its expected value approaches infinity

d. It ignores outliers

Estimating $\bar{x}$ uses 1 degree of freedom, resulting in $E[\hat{\sigma}^2_{MLE}] = \frac{n-1}{n}\sigma^2$ (Bessel's correction uses $n - 1$).

39 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

What is Bessel's correction to make the sample variance estimator unbiased?

a. **Multiply by $\frac{n}{n-1}$** ✓

b. Multiply by $\frac{n-1}{n}$

c. Add $\frac{1}{n}$

d. Subtract the mean

$s^2 = \frac{n}{n-1}\hat{\sigma}^2_{MLE} = \frac{1}{n-1}\sum(x_i - \bar{x})^2$.

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40 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 3.01 Gradient Descent and Optimization

What does it mean for an estimator $\hat{\theta}_n$ to be asymptotically consistent?

a. As sample size $n \to \infty$, $\hat{\theta}_n$ converges in probability to the true parameter θ ✓

b. The estimator has zero variance for all sample sizes

c. The estimator is unbiased for $n = 1$

d. The estimator follows a student-t distribution

Consistency is the property that with infinite data, the estimator converges to the ground truth parameter value.

41 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 25,\ \sigma^2 = 36$ ✓

b. $\mu = 36,\ \sigma^2 = 25$

c. $\mu = 25,\ \sigma^2 = 6$

d. $\mu = 6,\ \sigma^2 = 36$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 25$ and variance $\sigma^2 = 36$ (with standard deviation $\sigma = 6$).

42 [Challenging] • Target: Discrete Random Variables and Bernoulli Likelihood • Cross-Lesson: 6.01 Linear Regression and OLS

A Normal random variable has $\mu = 170$ and $\sigma = 15$. Find $P(148 < X < 190)$.

a. 0.5376

b. 0.8376 ✓

c. 0.7376

d. 0.6376

$z_1 = \frac{148 - 170}{15} = -1.467$, $z_2 = \frac{190 - 170}{15} = 1.333$. $P(148 < X < 190) = 0.8376$.

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43 [Challenging] • Target: Continuous Gaussian Normal Distributions • Cross-Lesson: 7.01 Logistic Regression and Classification

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. **n is large (n≥30) ✓**

c. σ is known

d. p<0.5

CLT: for large n, $\bar{x}$ is approximately Normal regardless of population shape.

44 [Challenging] • Target: The Likelihood Function and Log-Likelihood Formulation • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A Normal random variable has $\mu = 111$ and $\sigma = 10$. Find $P(102 < X < 116)$.

a. **0.5074 ✓**

b. 0.3074

c. 0.2074

d. 0.4074

$z_1 = \frac{102-111}{10} = -0.9$, $z_2 = \frac{116-111}{10} = 0.5$. $P(102 < X < 116) = 0.5074$.

45 [Challenging] • Target: Properties of Estimators: Bias and Consistency • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A Normal random variable has $\mu = 114$ and $\sigma = 10$. Find $P(100 < X < 119)$.

a. **0.6107 ✓**

b. 0.3107

c. 0.5107

d. 0.4107

$z_1 = \frac{100-114}{10} = -1.4$, $z_2 = \frac{119-114}{10} = 0.5$. $P(100 < X < 119) = 0.6107$.

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